

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

PRAGYA G8 – ACHIEVERS | MATHEMATICS
QUADRATIC EQUATIONS | LEVEL 1*Straightforward – one idea, one step.*

1. If $x = \sqrt{3 + 2\sqrt{2}}$, then $x^2 + \frac{1}{x^2}$ is equal to
(A) $2\sqrt{2}$ (B) 6
(C) 8 (D) 1
2. If $x = 3 - \sqrt{8}$, then $x^3 + \frac{1}{x^3}$ is equal to
(A) $6\sqrt{2}$ (B) 102
(C) 198 (D) 6
3. The roots of the equation $x^2 - 2\sqrt{2}x + 1 = 0$ are
(A) real and different (B) real and equal
(C) imaginary and different (D) rational and different
4. The roots of the equation $(b + c)x^2 - (a + b + c)x + a = 0$, where $a, b, c \in \mathbb{Q}$ and $b + c \neq a$, are
(A) imaginary and different (B) real and equal
(C) rational and different (D) irrational and different
5. The number of real solutions of $x - \frac{1}{x^2 - 4} = 2 - \frac{1}{x^2 - 4}$ is
(A) 4 (B) 0
(C) 1 (D) 2
6. The sum of the roots of the equation $(x + 3)^2 - 4|x + 3| + 3 = 0$ is
(A) 12 (B) 4
(C) -12 (D) -4
7. If α, β are the roots of $x^2 + px - q = 0$ and γ, δ are the roots of $x^2 + px + r = 0$, then $(\alpha - \gamma)(\alpha - \delta)$ is
(A) $p + r$ (B) $q - r$
(C) $q + r$ (D) $p - r$
8. If α, β are the roots of $x^2 - 5x + 6 = 0$, then the equation whose roots are $\alpha + 3$ and $\beta + 3$ is

(A) $(x - 3)^2 - 5(x - 3) + 6 = 0$

(B) both (A) and (B)

(C) $x^2 - 11x + 30 = 0$

(D) none of these

9. If α, β are the roots of $x^2 - 3x + 5 = 0$, then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is

(A) $x^2 - 4x + 4 = 0$

(B) $x^2 + 4x + 1 = 0$

(C) $x^2 + 2x + 3 = 0$

(D) $x^2 - 4x - 1 = 0$

10. The number of values of a for which $(a^2 - 3a + 2)x^2 + (a^2 - 5a + 6)x + a^2 - 4 = 0$ is an identity in x is

(A) 3

(B) 0

(C) 2

(D) 1

11. If $\alpha + \beta = 3$ and $\alpha^3 + \beta^3 = 7$, then α and β are the roots of

(A) $3x^2 + 9x + 7 = 0$

(B) $9x^2 - 27x + 20 = 0$

(C) $2x^2 - 6x + 15 = 0$

(D) none of these

12. The number of integral values of k for which the curve $y = x^2 + kx + 4$ touches the x -axis is

(A) 2

(B) 4

(C) 1

(D) 0

13. If the roots of $x^2 - bx + c = 0$ are two consecutive integers, then $b^2 - 4c$ equals

(A) 2

(B) 3

(C) -2

(D) 1

14. For the quadratic expression $y = x^2 - px + q$ with $p = 4$ and $q = 9$, the minimum value of the expression is

(A) 3

(B) 6

(C) 4

(D) 5

15. If $a, b, c > 0$, then the number of real roots of $ax^2 + b|x| + c = 0$ is

(A) 2

(B) 1

(C) 4

(D) none of these